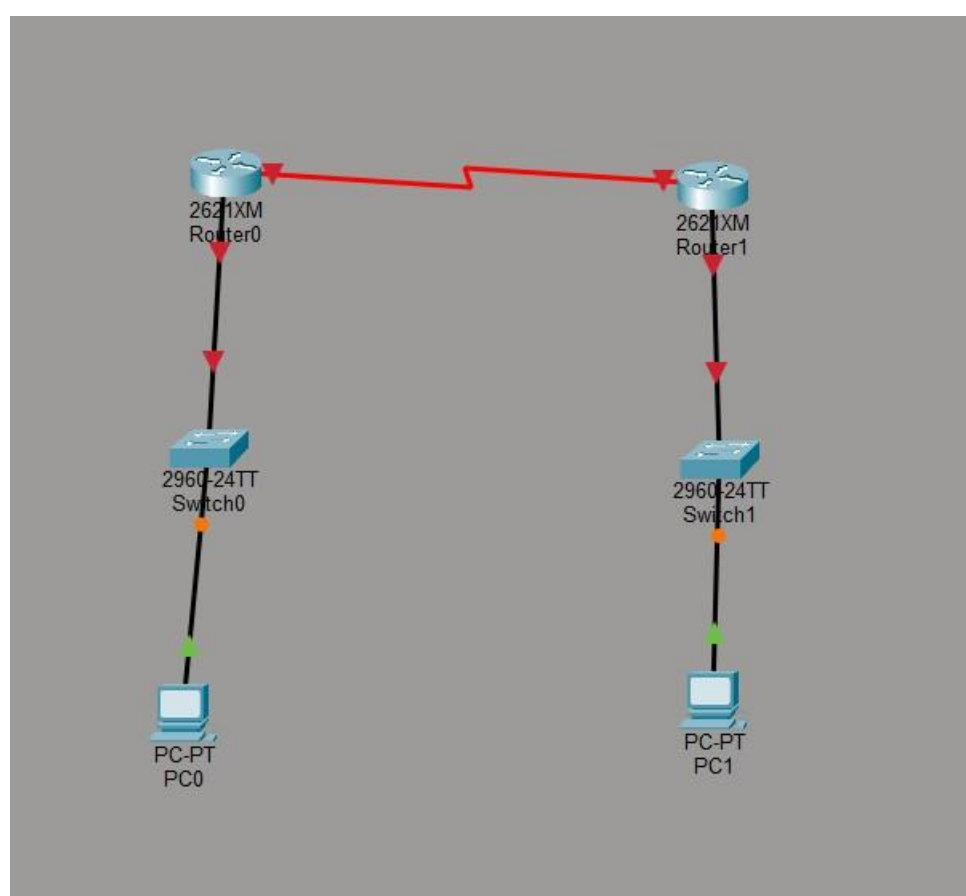


Task-1

- Create a simple routed network in Cisco Packet Tracer using two Cisco 2621XM routers connected by a serial link.
- Assign IPv4 addresses to the router LAN and serial interfaces.
- Configure static routes so the two LANs can communicate through the routers.
- Configure PC default gateways and verify connectivity using ping and show ip interface brief.

TOPOLOGY



3. IP Addressing Plan

Device	Interface	IP Address	Subnet Mask	Default Gateway
PC0	FastEthernet	192.168.1.10	255.255.255.0	192.168.1.1
Router0	FastEthernet0/0	192.168.1.1	255.255.255.0	-
Router0	Serial0/0	10.0.0.1	255.255.255.252	-
Router1	Serial0/0	10.0.0.2	255.255.255.252	-
Router1	FastEthernet0/0	192.168.2.1	255.255.255.0	-
PC1	FastEthernet	192.168.2.10	255.255.255.0	192.168.2.1

Router0

```
Router0
Physical Config CLI Attributes

Press RETURN to get started!

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
Router(config-if)#exit
Router(config)#ip route 192.168.2.0 255.255.255.0 10.0.0.2
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface                IP-Address      OK? Method Status      Protocol
FastEthernet0/0          192.168.1.1     YES manual up          up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0                 10.0.0.1        YES manual down        down
Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
.....
Success rate is 0 percent (0/5)

Router#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#ping 10.0.0.2

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.2, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 4/7/11 ms

Router#
```

Router1

Router1

PhysicalConfigCLIAttributes

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#ip address 192.168.1.0 255.255.255.0 10.0.0.1
^
% Invalid input detected at '^' marker.

Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

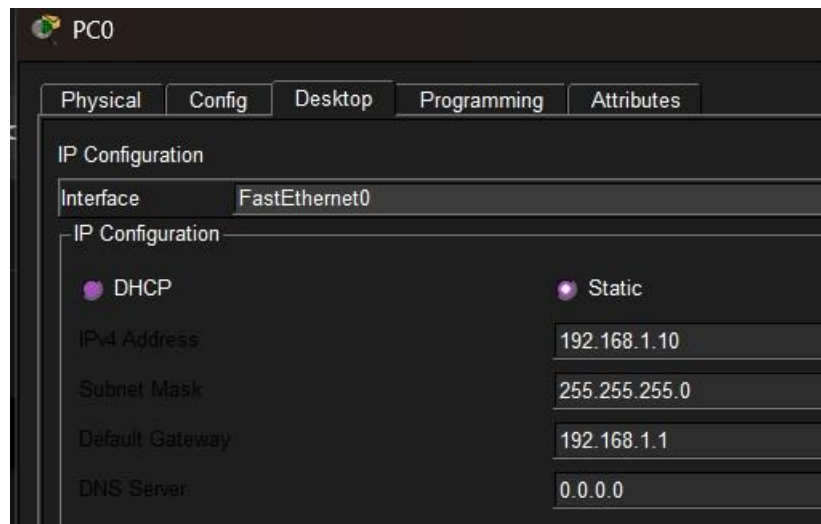
Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          192.168.2.1     YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0                 unassigned      YES unset  administratively down down
Router#show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          192.168.2.1     YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0                 unassigned      YES unset  administratively down down
Router#enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

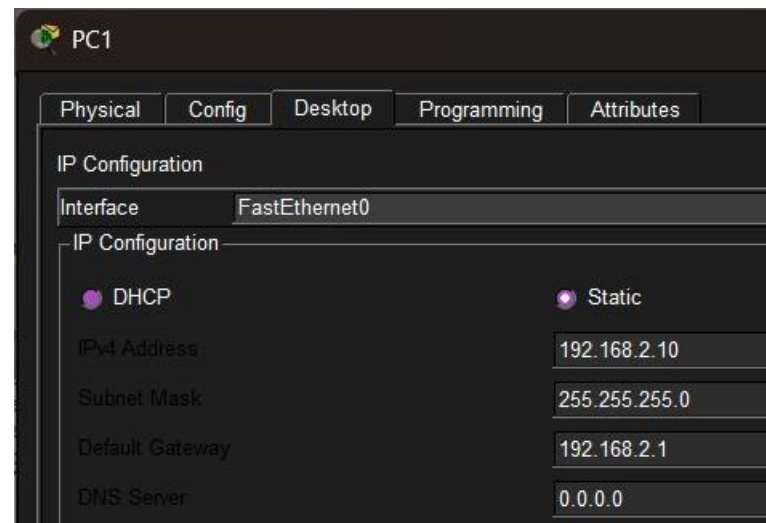
Router(config-if)#exit
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up
show ip interface brief
Interface                IP-Address      OK? Method Status          Protocol
FastEthernet0/0          192.168.2.1     YES manual up              up
FastEthernet0/1          unassigned      YES unset  administratively down down
Serial0/0                 10.0.0.2        YES manual up              up
Router#ping 10.0.0.1
```

PC0



PC1



TESTING

```
C:\>ping 192.168.2.10

Pinging 192.168.2.10 with 32 bytes of data:

Reply from 192.168.2.10: bytes=32 time=12ms TTL=126
Reply from 192.168.2.10: bytes=32 time=5ms TTL=126
Reply from 192.168.2.10: bytes=32 time=2ms TTL=126
Reply from 192.168.2.10: bytes=32 time=1ms TTL=126

Ping statistics for 192.168.2.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 12ms, Average = 5ms

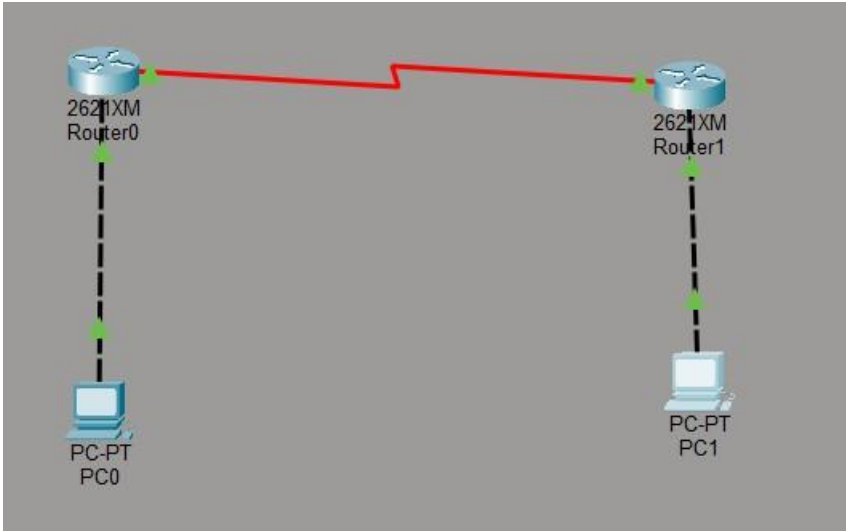
C:\>
```

Task-2

Q-On Router1 in Cisco Packet Tracer, configure a static route so it can reach a PC connected to Router2’s LAN network. Test connectivity using the ping command from Router1 to the PC.

The objective of this practical is to configure a static route on **Router1** so that it can communicate with a PC connected to the LAN of **Router0**. Connectivity is verified using the ping command.

TOPOLOGY



IP Addressing Table

Device	Interface	IP Address	Mask
Router0	Fa0/0	192.168.1.1	255.255.255.0
PC0	Fa0	192.168.1.10	255.255.255.0
Router0	Serial0/0	10.0.0.1	255.255.255.252
Router1	Serial0/0	10.0.0.2	255.255.255.252
Router1	Fa0/0	192.168.2.1	255.255.255.0
PC1	Fa0	192.168.2.10	255.255.255.0

Router0

Router0

PhysicalConfigCLIAttributes

Press RETURN to get started!

Router>enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.1.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#show ip interface brief
^
% Invalid input detected at '^' marker.

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#show ip interface brief
Interface IP-Address OK? Method Status Protocol
FastEthernet0/0 192.168.1.1 YES manual up up
FastEthernet0/1 unassigned YES unset administratively down down
Serial0/0 10.0.0.1 YES manual down down
Router#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router#ping 192.168.1.10

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 192.168.1.10, timeout is 2 seconds:
.!!!!
Success rate is 80 percent (4/5), round-trip min/avg/max = 0/0/0 ms

Router#

Router1

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
      ^
% Invalid input detected at '^' marker.

Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.2.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed state to up

Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#ping 10.0.0.1

Type escape sequence to abort.
Sending 5, 100-byte ICMP Echos to 10.0.0.1, timeout is 2 seconds:
!!!!!!
Success rate is 100 percent (5/5), round-trip min/avg/max = 1/4/7 ms
```

PC Configuration

PC0

IP Address: 192.168.1.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.1.1

PC1

IP Address: 192.168.2.10
Subnet Mask: 255.255.255.0
Default Gateway: 192.168.2.1

Static Route Configuration

```
Router#enable
Router#config t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#ip route 192.168.1.0 255.255.255.0 10.0.0.1
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#do wr
^
% Invalid input detected at '^' marker.

Router#show ip route
Codes: C - connected, S - static, I - IGRP, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

    10.0.0.0/24 is subnetted, 1 subnets
C       10.0.0.0 is directly connected, Serial0/0
S       192.168.1.0/24 [1/0] via 10.0.0.1
C       192.168.2.0/24 is directly connected, FastEthernet0/0

Router#
```

Connecting Testing

Router1# ping 10.0.0.1

The connection was successful.

Next, Router1 was used to test connectivity to PC0:

Router1# ping 192.168.1.10 successful output:

!!!!

Success rate is 100 percent (5/5)

This confirms that Router1 can successfully reach PC0 through Router0 using the configured static route.

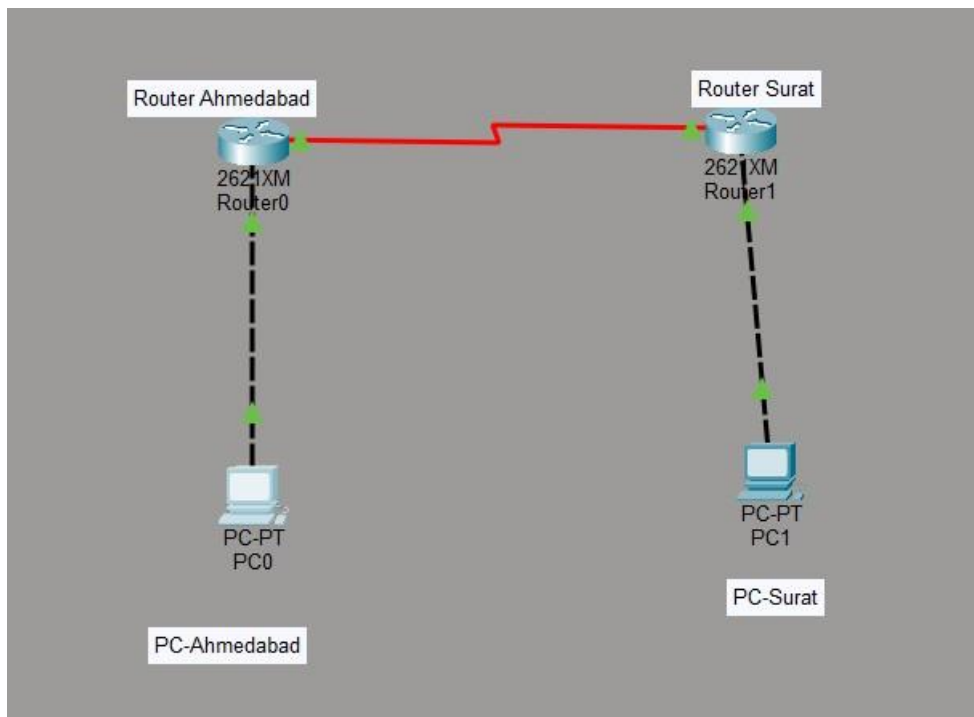
Task-3

Q-Simulate a Zomato-style delivery network: imagine Router1 represents an Ahmedabad branch and Router2 represents a Surat branch. Configure static routes so that devices in each city's LAN can communicate with each other through the routers.

The objective of this practical is to simulate a Zomato-style delivery network in Cisco Packet Tracer, where:

- Router1 represents the Ahmedabad branch . Router2 represents the Surat branch
- Each branch has its own LAN.
- Static routes are configured so devices in Ahmedabad and Surat can communicate with each other through the routers.
- Connectivity is verified using the ping command.

Network Topology



IP Addressing Table

Device	Interface	IP Address	Subnet Mask
Router1 Ahmedabad	FastEthernet0/0	192.168.10.1	255.255.255.0
PC-Ahmedabad	FastEthernet	192.168.10.10	255.255.255.0
Router1 Ahmedabad	Serial0/0	10.0.0.1	255.255.255.252
Router2 Surat	Serial0/0	10.0.0.2	255.255.255.252
Router2 Surat	FastEthernet0/0	192.168.20.1	255.255.255.0
PC-Surat	FastEthernet	192.168.20.10	255.255.255.0

Router0 – Ahmedabad Configuration

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.10.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.1 255.255.255.252
Router(config-if)#clock rate 64000
Router(config-if)#no shutdown

%LINK-5-CHANGED: Interface Serial0/0, changed state to down
Router(config-if)#exit
Router(config)#ip route 192.168.20.0 255.255.255.0 10.0.0.2
Router(config)#do wr
Building configuration...
[OK]
Router(config)#
```

Router1 – Surat Configuration

```
Router>enable
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#interface fastethernet 0/0
Router(config-if)#ip address 192.168.20.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface FastEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface FastEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#interface serial 0/0
Router(config-if)#ip address 10.0.0.2 255.255.255.252
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface Serial0/0, changed state to up

Router(config-if)#exit
Router(config)#ip route 19
%LINEPROTO-5-UPDOWN: Line protocol on Interface Serial0/0, changed s
Router(config)#ip route 192.168.10.0 255.255.255.0 10.0.0.1
Router(config)#do wr
Building configuration...
[OK]
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#
```

Test Connectivity

```
C:\>ping 192.168.20.10

Pinging 192.168.20.10 with 32 bytes of data:

Reply from 192.168.20.10: bytes=32 time=1ms TTL=126
Reply from 192.168.20.10: bytes=32 time=10ms TTL=126
Reply from 192.168.20.10: bytes=32 time=11ms TTL=126
Reply from 192.168.20.10: bytes=32 time=2ms TTL=126

Ping statistics for 192.168.20.10:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 1ms, Maximum = 11ms, Average = 6ms

C:\>
```

- ☐ **Ahmedabad LAN** represents Zomato's delivery operations and devices in Ahmedabad.
- ☐ **Surat LAN** represents Zomato's delivery operations and devices in Surat.
- ☐ **Router1** represents the Ahmedabad branch router.
- ☐ **Router2** represents the Surat branch router.
- ☐ The **serial connection** represents the communication link between the two cities.
- ☐ Static routes determine how traffic travels between the two city networks.

Task-4

Static Routing vs Dynamic Routing

Point	Static Routing – Advantage/Disadvantage	Real-life Example
1. Simple & Predictable	Advantage: Static routes are manually configured and follow a fixed path, making them easy to control in a small network.	In an IRCTC office network, a fixed route to the booking server can provide predictable communication.
2. Low Overhead	Advantage: Routers do not exchange routing updates, so less bandwidth and processing power are used.	A small Swiggy branch network could use static routes without spending bandwidth on routing updates.
3. No Automatic Failure Recovery	Disadvantage: If a configured route fails, the router normally will not automatically find another route.	If a network link used by IRCTC booking services fails, booking or payment updates could be interrupted until another route is configured.
4. Difficult to Scale	Disadvantage: Every route must be manually configured and updated, making large networks difficult to manage.	If Swiggy has many offices and delivery-related servers across cities, maintaining routes manually becomes time-consuming and can cause connectivity problems when the network changes.

Conclusion: Static routing is **simple and predictable for small networks**, but dynamic routing is generally more suitable for **large, changing networks** because it can learn and adapt to route changes automatically.